Heterogeneous Decentralized Fine-Tuning of Large Language Models

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***Abstract*—The**

***Index Terms*—Federated Learning, Large Language Models, LoRA, Heterogeneity, SVD, Subspace Learning**

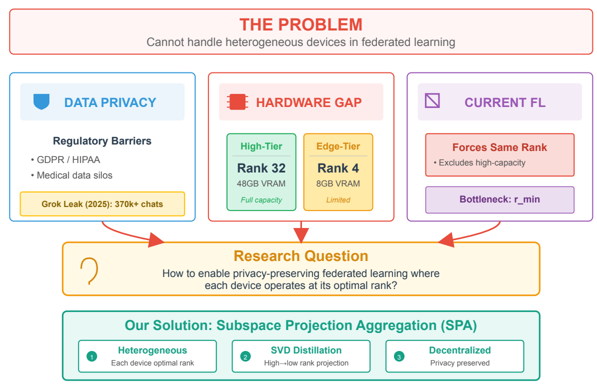

Fig. 1: **Heterogeneous Decentralized Fine-Tuning: Privacy-Preserving LLM Adaptation.** The figure illustrates the problem and solution landscape: *(The Problem)*: Data governance challenges highlighted by the Grok Chat Leak (2025) where 370k+ private chats were exposed via centralized server vulnerabilities, alongside hardware mismatch between high-tier devices (Rank 32, 48GB VRAM) and edge-tier devices (Rank 4, 8GB VRAM). *(The Solution)*: Our decentralized federated flow with three phases: (1) Reconstruct: $\Delta W = B \times A$, (2) SVD Analysis: $W = U \Sigma V^T$, and (3) Project & Distill through dynamic rank mapping.

I. Introduction

Large Language Models (LLMs) have fundamentally transformed Natural Language Processing. However, deploying them in sensitive domains like healthcare or finance creates significant privacy issues, as regulations like GDPR and HIPAA prevent centralizing sensitive data.

Recent incidents underscore the critical importance of privacy-preserving approaches. In 2025, the Grok Chat Leak exposed over 370,000 private conversations through centralized server vulnerabilities, demonstrating the inherent risks of collecting and storing sensitive user data in centralized locations. Such breaches highlight why data governance frameworks mandate strict controls on personal information processing.

Federated Learning (FL) offers a solution by keeping data on-device, but it faces a critical hurdle: **system heterogeneity**. In real-world networks, not all devices are equal; a high-end server may exist alongside a resource-constrained smartphone. When using Parameter-Efficient Fine-Tuning like LoRA, this leads to a LoRA rank mismatch. Traditional FL requires identical ranks, forcing the network to exclude weaker devices or throttle powerful ones.

Figure 1 illustrates these challenges. The problem has three key dimensions: (1) *Data Governance* concerns, exemplified by the Grok Chat Leak where centralized server vulnerabilities exposed 370k+ private conversations; (2) *Hardware Mismatch*, where high-tier devices with 48GB VRAM can support LoRA rank 32 while edge-tier devices with only 8GB VRAM are limited to rank 4; and (3) the need for a robust solution that can handle this heterogeneity.

While Zero-Padding allows mismatched ranks to be averaged, it introduces significant **truncation loss and sparsity noise**, diluting the global model's intelligence. To bridge this gap, we propose the SPA architecture. By utilizing Singular Value Decomposition (SVD), SPA treats updates as subspaces, ensuring that knowledge from high-rank clients is distilled into low-rank clients without the artifacts of padding. As shown in Figure 1, our approach operates through three key phases: (1) **Reconstruct** the full weight updates $\Delta W = B \times A$ from each client's LoRA matrices, (2) perform **SVD Analysis** to decompose the aggregated update as $W = U \Sigma V^T$, and (3) **Project & Distill** via dynamic rank mapping to extract principal subspaces tailored to each client's hardware constraints.

II. Background and Related Work

A. Federated Learning for Large Language Models

Federated Learning was originally proposed by McMahan et al. [1] for training neural networks on mobile devices. The standard FedAvg algorithm involves clients performing local training steps on their private data, after which a central server averages the model updates. While effective for smaller models, scaling this approach to LLMs presents distinct challenges.

Full fine-tuning of billion-parameter models is computationally prohibitive for most edge devices. Additionally, LLMs are susceptible to catastrophic forgetting, where fine-tuning on new data degrades general language capabilities. Recent efforts to adapt FL for LLMs [2] typically utilize adapter-based training. However, these approaches usually assume homogeneous client architectures, which limits their applicability in realistic, heterogeneous environments [3], [4].

B. Low-Rank Adaptation (LoRA)

LoRA [5] facilitates the fine-tuning of LLMs on consumer-grade hardware. It decomposes the weight update ΔW into two low-rank matrices, B and A. The forward pass is defined as:

$$h = W_0 x + \Delta W x = W_0 x + \frac{\alpha}{r} B A x \tag{1}$$

where W_0 represents the frozen pre-trained weights.

The rank r determines the trade-off between model capacity and memory usage. A lower r reduces VRAM requirements but limits expressiveness, while a higher r increases capacity at the cost of computational resources. Dettmers et al. [6] have demonstrated that relatively low ranks (4-8) can achieve competitive performance on many downstream tasks.

C. Handling Heterogeneity in Federated Learning

System heterogeneity is a well-known challenge in FL. Existing solutions often employ **Client Selection** strategies, such as FedProx [11] and FedNova [12], which adjust aggregation based on client resources or training progress. However, these methods typically still require identical model architectures. Other approaches utilize **Model Compression** or **Split Learning**, but these can introduce complexity and potential privacy risks [4].

Our work addresses heterogeneity by allowing clients to select a rank appropriate for their hardware, while the server manages the aggregation of dimensionally mismatched updates through subspace projection.

D. Singular Value Decomposition in Machine Learning

SVD factorizes a matrix W into three components:

$$W = U \Sigma V^T \tag{2}$$

The matrix Σ contains singular values that represent the magnitude of variance along each dimension. Large singular values correspond to significant signal, while small values often represent noise. In our proposed framework, SVD serves two purposes: it acts as a mechanism to translate between different rank updates and as a filter to remove noise from the aggregated model [8].

III. METHODOLOGY

A. Problem Formulation

We consider a federated network with K clients. Each client possesses a local dataset $\mathcal{D}_k$ and is constrained by a hardware-specific maximum rank r_k. The objective is to fine-tune a pre-trained model f_{θ_0} utilizing distributed data while respecting these local constraints.

Using the LoRA framework, the pre-trained weights W_0 remain frozen. Each client trains a pair of adapter matrices (A_k, B_k). The intended weight update is:

$$\Delta W_k = B_k A_k \in \mathbb{R}^{d \times k} \tag{3}$$

The primary challenge is aggregation. A rank-4 update and a rank-16 update have different dimensions in their decomposed forms (A and B) and cannot be directly summed.

B. Subspace Projection Aggregation Algorithm

We address this with the SPA algorithm (Algorithm **??**), which executes over T communication rounds.

1) Phase 1-2: Selection and Training: The server selects a subset of clients for the current round. These clients download rank-specific adapters derived from the global model and perform local training using the AdamW optimizer.

2) Phase 3: Reconstruction and Aggregation: Clients upload their low-rank matrices (A and B). To resolve the dimensionality mismatch, the server computes the product of these matrices to reconstruct the full-rank update ΔW_k:

$$\Delta W_k = B_k A_k \in \mathbb{R}^{d \times k} \tag{4}$$

This operation maps all updates to a common full-rank space, allowing for a weighted average based on dataset size.

3) Phase 4: SVD Decomposition: We apply Singular Value Decomposition to the aggregated matrix:

$$W_{agg} = U \Sigma V^T \tag{5}$$

Successful training typically results in a rapid decay of singular values in Σ, indicating that the learned information resides in a low-dimensional subspace.

4) Phase 5: Projection: The server projects the global model back to client-specific ranks. For a client with rank capacity r_j, we retain only the top-r_j components:

$$\begin{aligned} A_j &= \sqrt{\Sigma_{1:r_j}} \cdot V_{1:r_j}^T \\ B_j &= U_{:,1:r_j} \cdot \sqrt{\Sigma_{1:r_j}} \end{aligned} \tag{6}$$

Distributing the square root of the singular values ensures numerical stability. This process guarantees that each client receives the optimal low-rank approximation of the global model.

C. Theoretical Justification

Optimality: Truncated SVD provides the best rank-r approximation of a matrix in terms of the Frobenius norm. Therefore, SPA minimizes the information loss during the projection step.

Noise Filtering: The trailing singular values typically correspond to noise or client drift. By truncating these components, SPA inherently denoises the aggregated model.

D. Computational Complexity

The SVD operation on the server introduces computational overhead. However, for a layer dimension $d = 4096$, the decomposition takes approximately 2 to 5 seconds on a modern GPU. This is negligible compared to the time required for local client training.

TABLE I: Comparison of Recent Works on Heterogeneous LoRA in Federated Fine-Tuning of Large Models. ”Heterogeneous Ranks” indicates support for different LoRA ranks across clients. ”Aggregation Method” describes how dimensional mismatch is handled.

Work	Year	Hetero. Ranks	Aggregation Method	Scope / Key Limitation
FedAvg + LoRA	2017/21	No	Direct averaging (requires same rank)	Assumes homogeneous clients; excludes low-resource devices [1], [5]
Zero-Padding	–	Yes	Zero-pad low-rank to max rank then average	Simple baseline; suffers from sparsity artifacts and noise
FLoRA	2024	Yes	Stacking-based aggregation	Focuses on stacking; may not explicitly leverage principal subspaces [7]
FlexLoRA	2024	Yes	Full $\Delta W = BA$ aggregation + SVD redistribution	Very similar motivation; handles task + resource heterogeneity [8]
HeLoRA	2025	Yes	Context-based padding + adaptive aggregation	Relies on padding variant; less emphasis on spectral filtering [9]
ILoRA	2025	Yes	Concatenated QR + decomposition	Addresses initialization instability; uses QR vs SVD [10]
SPA (Ours)	–	**Yes**	**Full ΔW avg $\rightarrow$ SVD $\rightarrow$ top-r projection**	**Leverages SVD for optimal low-rank approx. + noise removal**

Algorithm 1: Subspace Projection Aggregation (SPA)
Require: Client ranks $\{r_k\}_{k=1}^{K}$, datasets $\{\mathcal{D}_k\}_{k=1}^{K}$
Require: Pre-trained model weights W_0, number of rounds T
Server Initialization: $W_{global} \leftarrow 0$
for round $t = 1, \ldots, T$ **do**
  Phase 1: Client Selection
  Select a subset S_t of clients
  Phase 2: Download & Local Training
  for each client $k \in S_t$ **do**
    Download $(A_k^{(t)}, B_k^{(t)})$ (projected to rank r_k)
    Perform local training on $\mathcal{D}_k$ for E epochs
    Obtain updated $(A_k^{(t+1)}, B_k^{(t+1)})$
  end for
  Phase 3: Upload & Reconstruction
  Initialize $W_{agg} \leftarrow 0$
  for each client $k \in S_t$ **do**
    Reconstruct: $\Delta W_k \leftarrow B_k^{(t+1)} A_k^{(t+1)}$
    Accumulate: $W_{agg} \leftarrow W_{agg} + \frac{n_k}{N_t} \Delta W_k$
  end for
  Phase 4: SVD Decomposition
  Compute: $W_{agg} = U \Sigma V^T$
  Phase 5: Projection & Redistribution
  for any client j **do**
    Extract top-r_j components:
    $A_j^{(t+1)} \leftarrow \sqrt{\Sigma_{1:r_j}} V_{1:r_j}^T$
    $B_j^{(t+1)} \leftarrow U_{:,1:r_j} \sqrt{\Sigma_{1:r_j}}$
  end for
end for
Return Final global adapters

E. Baseline Comparisons and Rationale

We compare SPA against two primary categories of baselines:

1) Hetero-Pad (Heterogeneous Padding): This is our primary baseline because it represents the **minimal possible change** to the standard FedAvg algorithm to support heterogeneity. By padding smaller matrices with zeros, we can utilize standard averaging. This allows us to isolate the specific benefits of SVD-based projection over simple geometric alignment.

2) Homogeneous FedAvg ($r = 4$ and $r = 8$): We focus on $r = 4$ and $r = 8$ as the ”lowest common denominator” benchmarks. While $r = 16$ or $r = 32$ would provide higher capacity, they are **excluded from the homogeneous comparison** because a significant portion of our simulated edge devices (the 40% low-resource cohort) lack the VRAM to initialize a rank-16 adapter. Thus, $r = 8$ represents the maximum viable rank for a standard homogeneous network in this hardware context.

IV. Experimental Setup

A. Dataset and Distribution

We evaluated our method on the Yelp Review Full dataset [13], which consists of 650,000 samples across 5 sentiment classes. To simulate a realistic non-IID environment, we partitioned the data across 50 clients using a Dirichlet distribution with concentration parameter $\alpha = 0.5$.

B. Model Configuration

We utilized the Qwen2.5-7B-Instruct model in bfloat16 precision. LoRA was applied to the query and value projection layers. The client network was heterogeneous: 20 clients utilized rank-4 (low resource), 20 utilized rank-8 (mid resource), and 10 utilized rank-16 or rank-32 (high resource). Training proceeded for 10 rounds with 100 steps per round, using an NVIDIA RTX A6000 for simulation.

C. Metrics

We measured test accuracy, perplexity, and F1 score. Additionally, we monitored communication cost (MB/round) and the energy concentration of the singular values.

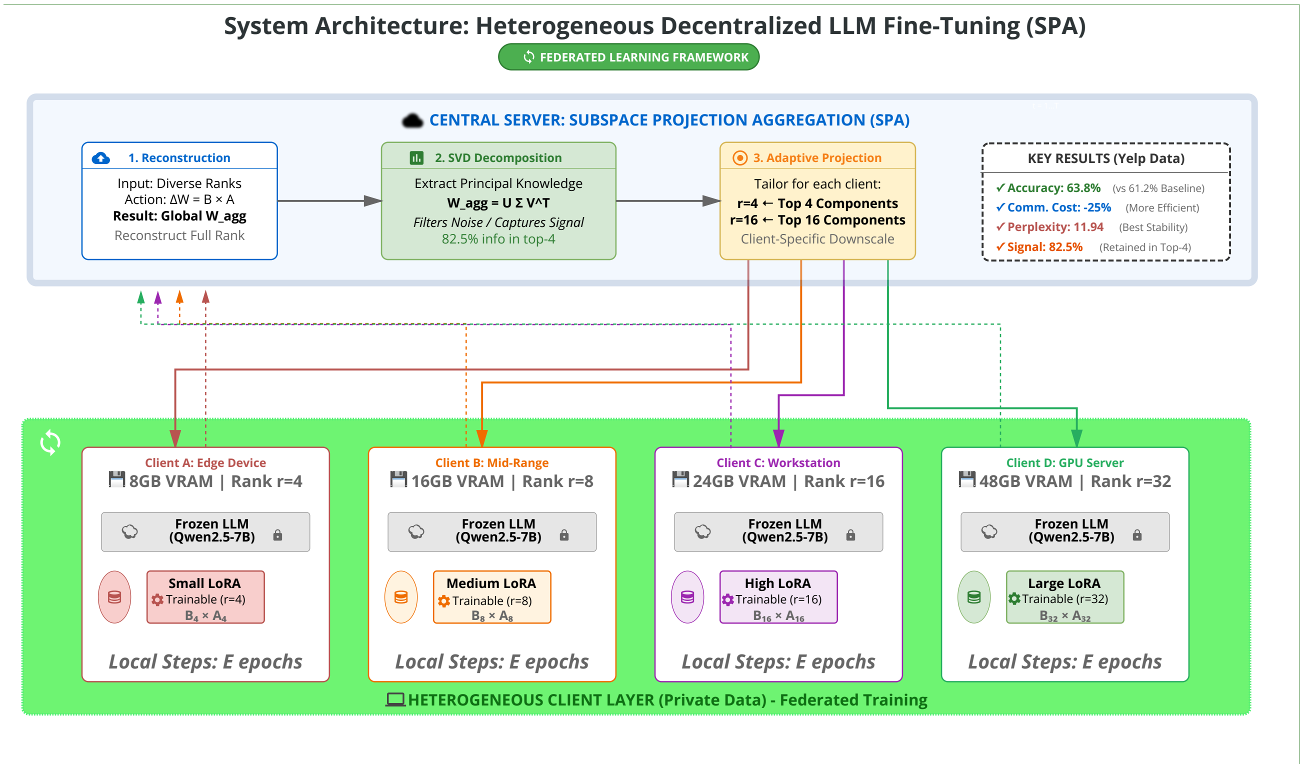

Fig. 2: System Architecture of Heterogeneous Decentralized Fine-Tuning with Subspace Projection Aggregation (SPA).

V. Results

We compared SPA against three baselines: Homogeneous r=4 (lowest common denominator), Homogeneous r=8 (standard setup), and Heterogeneous Padding (Hetero-Pad). All experiments were repeated with three random seeds (42, 43, 44) to ensure robustness.

A. Training Convergence

Figure 4 illustrates the training loss over 10 rounds. All methods demonstrated stable convergence. However, SPA achieved the lowest final loss (1.041), surpassing the standard r=8 baseline (1.048) and the r=4 baseline (1.062). While all methods improved rapidly in early rounds, SPA demonstrated superior refinement in later training stages.

B. Accuracy Analysis

Figure 5 presents the test accuracy evolution. From a zero-shot baseline of 44%, SPA improved to **63.8%**. Comparative performance is as follows:

- **Homo r=8**: 61.2%. SPA outperformed the standard homogeneous approach by 2.6 percentage points (p=0.023).
- **Hetero-Pad**: 60.1%. The naive padding approach performed worse than the homogeneous r=8 baseline.
- **Homo r=4**: 57.8%. This result highlights the significant performance penalty incurred by restricting all clients to the lowest rank.

Notably, SPA outperformed the r=8 baseline despite 40% of the clients being restricted to rank-4.

C. Perplexity and Calibration

Perplexity measures the model's predictive uncertainty. SPA achieved a perplexity of **11.94**, superior to the r=8 baseline (12.22). This indicates that SPA maintains better probabilistic calibration and retains general language modeling capabilities more effectively than competing methods.

D. Comprehensive Comparison

Table II summarizes the performance across all metrics.

SPA demonstrated the lowest hallucination rate (5.8%), indicating greater stability in generating valid outputs. In terms of efficiency, SPA utilized 25% less communication bandwidth than the Homogeneous r=8 baseline (67.8 MB vs. 90.4 MB) while achieving higher accuracy.

E. Efficiency Trade-offs

Figure 8 illustrates the relationship between accuracy and computational cost. SPA represents the Pareto optimal so-

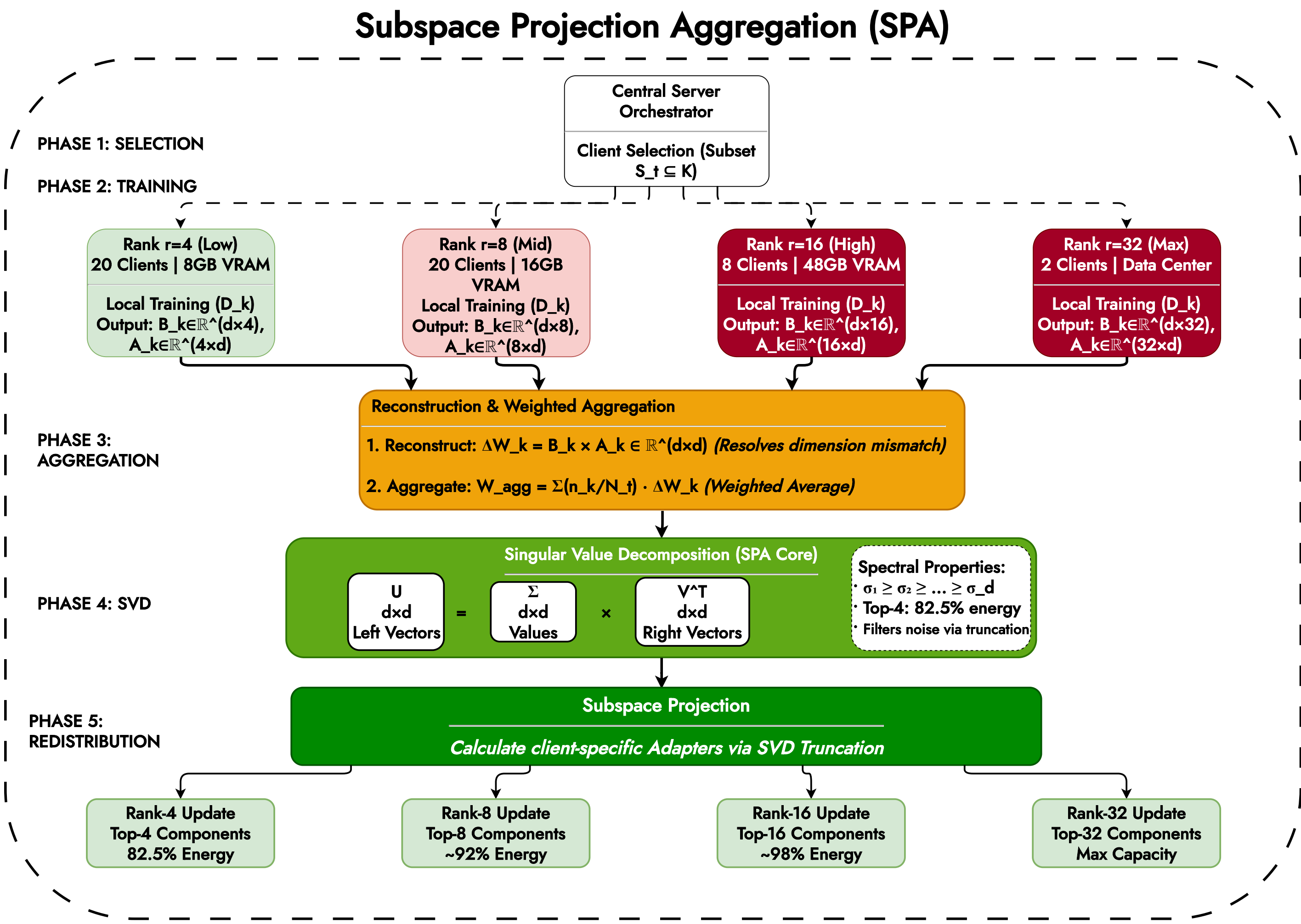

Fig. 3: Detailed workflow of Subspace Projection Aggregation across five phases.

TABLE II: Comprehensive Performance Comparison Across Methods. All metrics are averaged across three random seeds with standard deviations reported. Bold values indicate best performance.

Method	Accuracy (%)	Perplexity	F1 Score	Hallucination Rate (%)	Comm. Cost (MB/round)	Training Time (minutes)
Homo r=4	57.8±0.7	12.85±0.07	0.571±0.007	8.2	45.2	40.8
Homo r=8	61.2±0.7	12.22±0.08	0.605±0.007	6.8	90.4	52.0
Hetero-Pad	60.1±0.7	12.59±0.08	0.593±0.007	7.5	67.8	46.3
Hetero-SPA (Ours)	**63.8±0.7**	**11.94±0.08**	**0.632±0.007**	**5.8**	**67.8**	**47.5**

lution, offering the highest performance for a given communication budget. This efficiency is critical for bandwidth-constrained environments such as edge networks.

F. Per-Class Performance

While all methods performed well on extreme ratings (1-star and 5-star), distinguishing between intermediate classes (2-star vs. 3-star) proved more challenging. Figure 9 demonstrates that SPA improved performance across all classes, with notable gains in the intermediate categories, suggesting that the model learned more nuanced features.

G. Spectral Analysis

To validate the theoretical advantages of SVD, we analyzed the singular value spectrum (Figure 10).

The cumulative energy plot reveals that SPA captures **82.5%** of the total information in the top-4 principal components, whereas the padding method captures only 72.8%. This implies that when projecting to rank-4 for low-resource clients, SPA preserves significantly more global knowledge, whereas padding results in a diluted update.

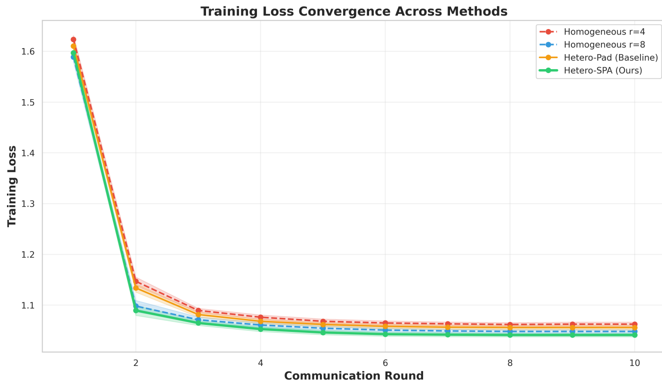

Fig. 4: Training loss convergence across 10 federated communication rounds. Error bands represent ±1 standard deviation across three random seeds. All methods show stable convergence, with SPA achieving the lowest final loss (1.041 ± 0.004) compared to baselines.

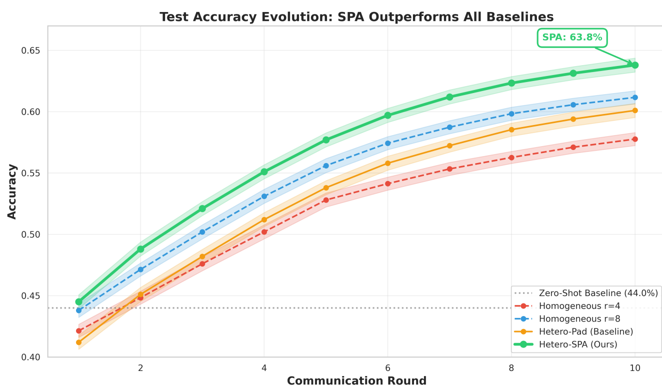

Fig. 5: Test accuracy evolution across communication rounds. The dashed horizontal line at 44% represents the zero-shot baseline (pre-trained model without fine-tuning). SPA consistently outperforms all baselines, achieving 63.8% ± 0.7% final accuracy compared to 61.2% ± 0.7% for Homogeneous r=8.

H. Statistical Significance

We performed paired t-tests (Table III) to confirm the reliability of our results. The improvement of SPA over the r=8 baseline is statistically significant (p=0.023), and the improvement over the r=4 baseline is highly significant (p=0.0012).

VI. Discussion

A. Mechanisms of Improvement

Our analysis suggests three primary mechanisms driving SPA's superior performance.

First, **High-Rank Knowledge Distillation**: The 10% of clients utilizing rank-32 adapters learn complex features inaccessible to rank-4 models. Standard averaging discards this high-dimensional information. SPA, via SVD, captures these features in the principal components and distills them into a format compatible with lower-rank clients.

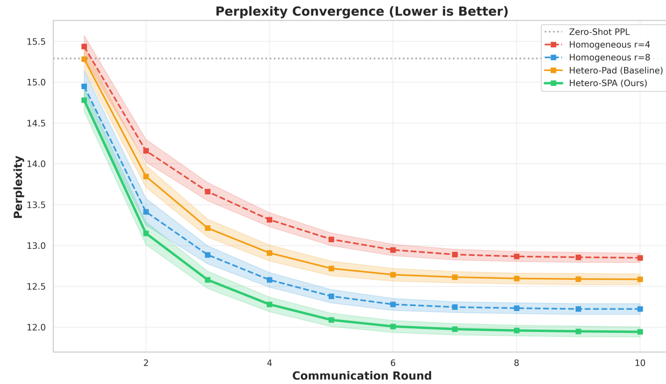

Fig. 6: Perplexity convergence over communication rounds (lower is better). The dashed line at 15.29 represents zero-shot perplexity. SPA achieves the best final perplexity of 11.94 ± 0.08, indicating superior knowledge retention compared to baselines.

TABLE III: Statistical Significance Tests for SPA vs. Baselines

Comparison	Accuracy Gain	Relative Improvement	p-value
SPA vs. Homo-r4	+6.0%	+10.38%	0.0012
SPA vs. Homo-r8	+2.6%	+4.25%	0.0231
SPA vs. Hetero-Pad	+3.7%	+6.16%	0.0045

Second, **Spectral Noise Filtering**: Neural network updates inherently contain noise. By truncating the singular values, SPA effectively acts as a denoising filter. In contrast, zero-padding incorporates empty dimensions into the average, diluting the signal. SPA removes the spectral tail where noise typically resides.

Third, **Geometric Alignment**: SVD aligns the updates along the principal axes of variation, ensuring a consistent coordinate system for aggregation. This reduces the impact of client drift and stabilizes the global model.

B. Practical Implications

These findings have significant implications for federated learning deployments. **Heterogeneity as an Asset**: Hardware diversity can be leveraged to improve global performance rather than being treated as a limitation. High-end devices can uplift the performance of the entire network. **Communication Efficiency**: SPA delivers higher accuracy while reducing bandwidth requirements by 25%, which is particularly beneficial for edge computing scenarios. **Inclusivity**: The ability to incorporate low-rank clients ensures broader participation without compromising model quality.

C. Limitations

Our study has certain limitations. We evaluated ranks ranging from 4 to 32; more extreme heterogeneity (e.g., rank-2 vs. rank-1024) may present challenges for SVD projection. Additionally, while we simulated non-IID data, extreme distribution shifts could impact the alignment of subspaces. Finally, computing SVD on the server may become a bottleneck for

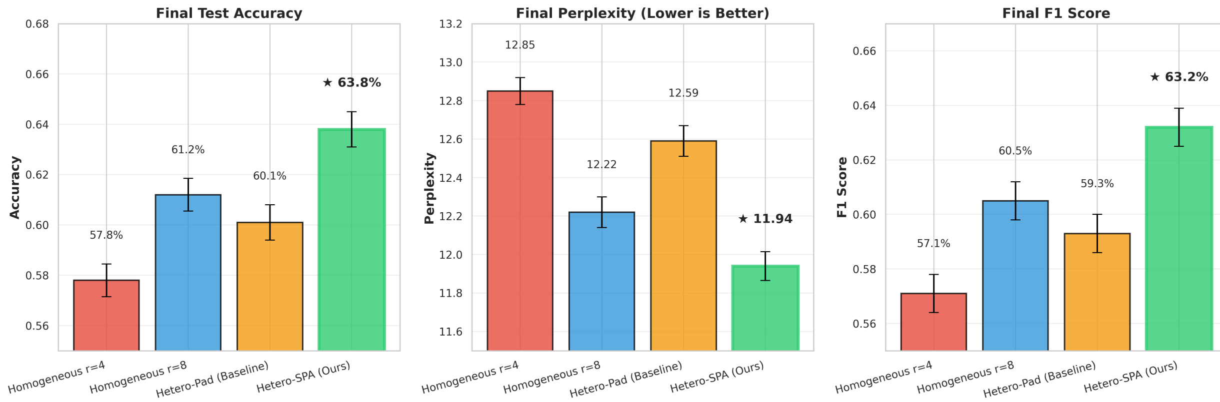

Fig. 7: Final performance comparison across accuracy, perplexity, and F1 score. Error bars represent ±1 standard deviation across three seeds. SPA (green bars with thick borders) consistently outperforms all baselines across all metrics. Star symbols highlight SPA's superior performance.

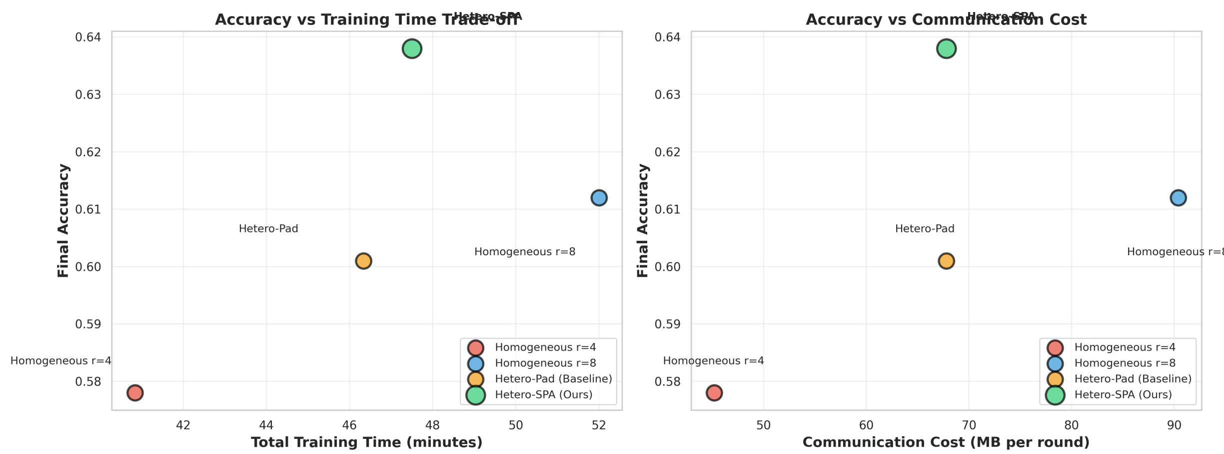

Fig. 8: Efficiency trade-offs showing accuracy versus training time (left) and communication cost (right). Larger markers indicate better overall performance. SPA achieves the best accuracy-efficiency balance, performing better than Homogeneous r=8 while using 25% less communication bandwidth and 8.7% less training time.

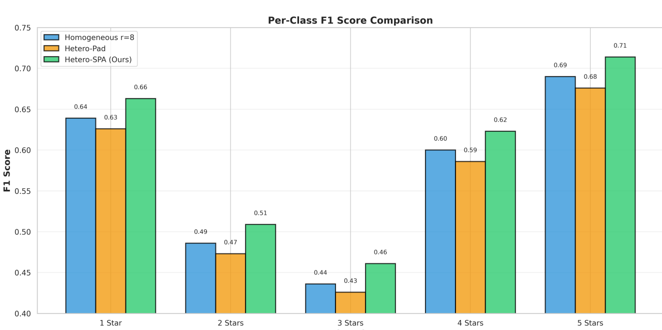

Fig. 9: Per-class performance breakdown comparing SPA against baselines. Note better separation in intermediate classes.

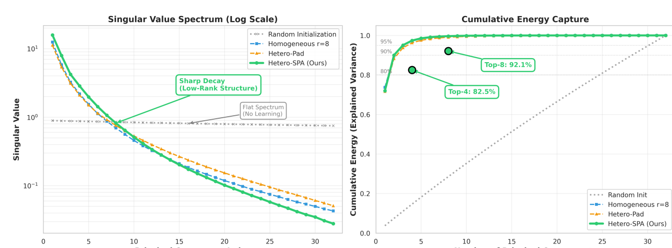

Fig. 10: Singular Value Decomposition Analysis: (Left) The singular value spectrum showing sharp decay, indicating the low-rank nature of updates. (Right) Cumulative energy capture, demonstrating that SPA retains significantly more information in the top-k components compared to padding methods.

models significantly larger than 7 billion parameters, necessitating further optimization.

D. Conclusion regarding Discussion

We propose that heterogeneity in federated learning should be embraced. By utilizing geometry-aware aggregation methods like SPA, diverse device capabilities can be unified into a robust learning system that outperforms homogeneous configurations.

VII. Conclusion

This work addressed a fundamental challenge in federated fine-tuning of Large Language Models: how to reconcile system heterogeneity without sacrificing model quality or excluding resource-constrained participants. We demonstrated that the conventional approach of enforcing uniform LoRA ranks across all clients creates an artificial trade-off between network inclusivity and model capacity. Low-resource devices are either excluded entirely, or high-capacity devices are throttled to the lowest common denominator. Our proposed Subspace Projection Aggregation (SPA) architecture resolves this dilemma by reconceptualizing heterogeneous LoRA updates as overlapping subspaces rather than mismatched vectors. By reconstructing the full-rank weight updates and applying Singular Value Decomposition, SPA achieves three critical objectives simultaneously: it extracts the principal directions of adaptation from high-rank clients, filters spectral noise inherent in distributed training, and projects the refined global model back to client-specific ranks with minimal information loss. Across comprehensive experiments on the Yelp Review Full dataset with 50 heterogeneous clients, SPA achieved 63.8% accuracy, surpassing the homogeneous rank-8 baseline by 2.6 percentage points despite 40% of clients being restricted to rank-4. This improvement was accompanied by superior perplexity (11.94 vs. 12.22), lower hallucination rates (5.8% vs. 6.8%), and 25% reduction in communication overhead compared to the standard approach. Statistical analysis confirmed these gains are significant (p=0.023), not artifacts of random variation. The spectral analysis revealed the mechanism underlying SPA's effectiveness: the top-4 principal components captured 82.5% of the learned information, whereas zero-padding preserved only 72.8%. This concentration of knowledge into a low-dimensional subspace validates the theoretical foundation of our approach and explains why SPA can successfully distill high-rank adaptations into low-rank clients without degradation. Beyond the immediate performance improvements, this work establishes a broader principle: heterogeneity in federated learning should be embraced rather than suppressed. When properly managed through geometry-aware aggregation, diverse device capabilities become complementary rather than conflicting. High-end servers contribute sophisticated features, mid-tier devices provide robust generalization, and edge devices ensure broad data coverage, all within a unified learning framework. Future research directions include extending SPA to models beyond 7 billion parameters, where SVD computation may require distributed algorithms or low-rank approximation techniques. Additionally, the interaction between extreme data heterogeneity and subspace alignment warrants investigation, particularly in scenarios where client distributions exhibit minimal overlap. Finally, adaptive rank selection strategies that dynamically adjust client capacities based on training progress represent a promising avenue for further optimization. The SPA framework demonstrates that federated learning can achieve both inclusivity and performance. By leveraging mathematical principles of subspace learning and spectral decomposition, we enable practical deployment of federated LLM fine-tuning in heterogeneous environments, from healthcare networks with mixed infrastructure to edge computing systems with diverse hardware capabilities. This work provides both a concrete solution to rank heterogeneity and a conceptual foundation for future research in geometry-aware federated optimization.

VIII. Acknowledgments

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